

Ran Wang

Applied Empirical Research | Urban & Regional Economics | High-Skilled Mobility and Innovation
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RESEARCH PROFILE

I am a quantitative empirical researcher trained in economic geography and regional innovation, with a substantive focus on high-skilled mobility, regional innovation systems, urban networks, and the spatial allocation of scientific talent. My work builds longitudinal micro-city datasets from career histories, publication records, institutional affiliations, and city-level indicators, and uses event-level destination-choice models, spatial and network analysis, and measurement-validity audits to study how scientific mobility and regional opportunity structures interact.

EDUCATION

East China Normal University, School of Geographic Sciences

Sep 2023 – Jun 2026

M.Sc. in Human Geography; research field: Economic Geography and Regional Innovation; GPA: 3.59/4.0; admitted by recommendation

- Master's thesis: Intercity Mobility Network of Top Global Biomedical Scientists: A Dual Perspective of Whole and Ego Networks.
- Selected coursework: Innovation Economics, Relational Economic Geography, Quantitative and Interdisciplinary Research Methods.

Northeast Normal University, School of Geographical Sciences

Aug 2019 – Jun 2023

B.Sc. in Human Geography and Urban-Rural Planning

- Coursework and training in quantitative geography, GIS spatial analysis, urban geography, regional analysis and planning, and urban planning methods.

RESEARCH INTERESTS

Urban and Regional Economics; Labor and High-Skilled Mobility; Regional Innovation; Science of Science; Innovation Networks; Spatial and Relational Data; Applied Empirical Research.

PUBLICATIONS AND MANUSCRIPTS

Wang, Ran. "Spatial Structure and Determinants of Global Top Biomedical Scientists' Mobility from an Urban Network Perspective."

Accepted

World Regional Studies, first author

- Constructed career-history and intercity mobility data from the global top 2% scientists list; applied social network analysis and a Heckman two-step model to study mobility-tie formation and mobility intensity across career stages.

"Pre-move Publication-Location Exposure and Destination Choice among Elite Biomedical Scientists." *Manuscript in progress*

- Reframes elite scientists' observed intercity moves as event-level destination choices; develops origin-anchored pre-move publication-location exposure as a measurable destination-visibility construct.

"Intercity Mobility Network of Top Global Biomedical Scientists: A Dual Perspective of Whole and Ego Networks." *Master's thesis*

- Integrates city-level mobility networks with scientist-level ego-network analysis using a 1970–2023 longitudinal database of 500 scientists and a 40-scientist ego-network subsample.

RESEARCH EXPERIENCE

Destination Choice, Publication-Location Exposure, and Scientific Mobility

2026 – Present

- Built event-specific candidate-city choice sets for observed intercity mobility events and constructed temporally ordered, origin-anchored pre-move publication-location exposure measures.
- Estimated event-stratified conditional-logit models and conducted timing, coverage, observability, and candidate-set diagnostics to assess the association between pre-move publication-location exposure and destination choice.
- Conducted strict timing, chosen/non-chosen coverage, complete-choice-set, observability-restricted subset, city-identity, and sample-overlap checks to evaluate candidate-set validity, differential missingness, and interpretation boundaries.
- Explicitly distinguished paper-level affiliation-city exposure from direct interpersonal collaboration and verified career mobility, reducing the risk of construct conflation.

Global Mobility and Collaboration Networks of Elite Biomedical Scientists

2024 – Present

- Assembled and harmonized a 1970–2023 longitudinal database of 500 elite biomedical scientists, integrating career histories, institutional appointments, publication locations, collaboration records, mobility events, and city attributes.
- Constructed scientist-institution-city linkages, directed weighted mobility edge lists, and multi-period city networks; analyzed network structure using centrality, core-periphery structure, K-core, valued ERGMs/WERGMs, and ego-network measures.
- Measured scientist-level collaboration embeddedness across career stages using constraint, effective size, Shannon entropy, and selected ego-network case studies; connected individual career trajectories with city-level mobility structures.
- Developed Python and R workflows for multi-source cleaning, entity standardization, temporal auditing, network construction, and analytical documentation.

Author–Institution Linkage and Geocoding Feasibility Audit

2026

- Developed a reproducible synthetic audit framework for author–institution–city linkage, including schema validation, deterministic institution normalization, identifier linkage, city–country validation, and record-level quality assurance.
- Documented measurement boundaries between publication affiliations, employment locations, and verified career mobility, while separating public synthetic materials from restricted research workflows.

Technological and Brand Innovation among Leading Global R&D Firms

2026 – Present

- Organizing R&D expenditures, patent portfolios, trademark portfolios, industry classifications, and country information for the world's top 2,000 corporate R&D investors into a firm-year analytical framework.
- Current work emphasizes data harmonization, defensible innovation measurement, and clear separation between completed data assets and proposed indicators.

ADDITIONAL QUANTITATIVE AND SPATIAL RESEARCH

Urban Resilience Measurement from Daily Nighttime Lights

2021 – 2023

- Constructed high-frequency urban radiance time series from VIIRS-DNB daily nighttime-light data and operationalized a Fisher Information Light indicator within an adaptive-cycle framework.
- Applied dynamic time warping and K-Means clustering to compare resilience trajectories across 48 cities in the Beijing–Tianjin–Hebei, Yangtze River Delta, and Pearl River Delta urban agglomerations.

Global Trade Network Evolution during COVID-19

Undergraduate research

- Built 2019–2021 global trade networks from UN Comtrade data and compared centrality, core structure, and spatial organization before and during the pandemic.

RELEVANT RESEARCH AND POLICY EXPERIENCE

Research Intern, China Development Institute, Shenzhen

- Supported policy-oriented research on port planning and regional development through literature and policy review, data organization, case collection, and report preparation.

Planning and Regional Development Research Projects

- Contributed to territorial, industrial, and rural-revitalization planning through spatial diagnosis, GIS mapping, policy-text preparation, and case analysis.

QUANTITATIVE MODELING EXPERIENCE

21st “Huawei Cup” China Graduate Mathematical Modeling Competition: Data-driven Modeling of Magnetic Core Loss

2024

- Compared Lasso, recursive feature elimination, and random-forest-based feature selection; built SVM classifiers with grid search and K-fold cross-validation.
- Log-linearized the Steinmetz equation and estimated parameters using OLS; contributed to sensitivity analysis and model interpretation.

SCHOLARSHIPS AND HONORS

- Outstanding Student Award, East China Normal University.
- Academic Scholarship, East China Normal University.
- Third-Class Scholarship, Northeast Normal University.
- Innovation and Entrepreneurship Scholarship, Northeast Normal University.
- National Undergraduate Innovation Training Program, Excellent Completion / First Place, Project Lead.

TECHNICAL SKILLS

- Empirical methods: conditional logit, OLS, Heckman two-step model, event-level candidate-set construction, cluster-robust inference, timing and coverage diagnostics, and sample-selection checks.
- Network and spatial methods: social network analysis, valued ERGMs/WERGMs, ego-network analysis, GIS spatial analysis, centrality, core–periphery structure, K-core, structural holes, and spatial visualization.
- Programming and tools: Python (Pandas, NumPy, NetworkX), R, Stata, ArcGIS, ENVI, Gephi, UCINET, Scimago Graphica.
- Languages: Mandarin Chinese (native); English academic reading and writing; CET-6.